Tech For Good 2024(Wellbeing) Proposal

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Part 1: Introduction for Aquaponics

Aquaponics in a sustainable and innovative farming method that combines both techniques of aquaculture (the cultivation of fish) and hydroponics (growing plants in nutrient-rich water) in a symbiotic environment. This integrated system creates a closed-loop ecosystem where the waste produced by the fish provides nutrients for the plants, and the plants, in turn, filter and purify the water for the fish. Aquaponics offers a holistic and resource-efficient approach to food production, utilizing the natural interactions between aquatic and plant life.

Aquaponics provides several advantages over traditional farming methods. The system uses significantly less water compared to conventional soil-based agriculture and it also eliminates the need for chemical fertilizers, as the fish waste serves as an organic and nutrient-rich alternative. Additionally, aquaponics allows for year-round cultivation, making it a reliable and sustainable method for fresh produce regardless of external weather conditions.

Taking it to another level, vertical farming is an agricultural practice that involves growing crops in vertically layers, which maximises the use of available space to produce the maximum crop wield. Vertical farming usually uses technologies such as aeroponics, and aquaponics, which allow for soil-less cultivation and precise control over water usage and environmental conditions.

Mashing up vertical farming and aquaponics contributes to a better control of space utilization and eliminates the need for chemical fertilizers, as the fish waste serves as an organic and nutrient-rich alternative. This makes a highly productive and space-efficient system that addresses both the challenges of traditional agriculture and the limitations of space in urban environments.

This is considered as an eco-friendly and scalable solution for producing both fish and vegetables in a symbiotic and closed-loop system.

Part 2: Problem Statement

Climate change has been the biggest problem faced by humanity ever since mankind started the industrial evolution. With us harnessing immeasurable quantities of non-renewable energy, we're constantly putting enormous amount of stress on our surrounding environment. According to the research done by European Commission from year 2011 - 2020, human activities that have the biggest influence on climate change are none other than burning fossil fuels, cutting down forests, farming livestock, and fertilizers containing nitrogen. These unending pursuits for technological greatness had caused the concentration of greenhouse gases in particularly carbon dioxide, methane, nitrous oxide and fluorinated gases to have risen to 48% above its pre-industrial level (before 1750). These greenhouse gases have been accelerating the melting of polar ice caps, increasing the frequency of extreme weather events and caused massive degradation of biodiversity on earth.

Moreover, human activities had also produced toxins and pollutants that exert enormous pressure on our ecosystem, in particularly our agricultural practices and energy generation. Humanity while striving for higher efficiency at production had often omitted the importance of living in harmony with our nature, instead we discharged waste directly into the environment, causing ripple effect that progressively intensifies over time. To be specifically, our artificially enriched fertilizers had leaked into rivers and oceans, causing abnormal nutrients to accumulate and lead to algae bloom or poisoning of aquatic life. Our energy producing factories such as hydrocarbon power plants had also released sulphur dioxides, carbon monoxides, particulate matter (PM), and even neural toxins like mercury and other types of heavy metals into the air which causes severe health impacts on humans, degrading the air quality even further. These man-made pollutants had been a major source of carcinogenic compounds according to Cancer Research UK since the industrial revolution and the culprit behind illnesses such as asthma, chronic obstructive pulmonary disease (COPD), heart disease and dementia. It also correlates to an increase in 10% for lung cancer due to the PM that causes inflamation of lung tissues and ultimately cancer.

Addressing thses complex and intertwined climate and pollution challenges requires a self sustaining but ecologically friendly agricultural methods that remodel the existing technology to provide sustainability and health over production efficiency.

Part 3: Aim and Objectives

The aim and objectives of our project is to construct a true self-sustaining agriculture with the help of current renewable energy technologies and mutually coexisting environment relations between plants and animals to provide a high quality, zero pollution farming. By using the aquaponic agricultural method, we arrived at a symbiotic relationship between small crops and fish, which is a cycle of nutrients between organisms, where the plants thrive off of the fishes' excrement while the fishes gets fed by the plants' fruits or seeds. The method had been proven to be extremely efficient in terms of utilizing the resources already inside the system to its maximum potential, while minimizing any waste produced.

Our team had also decided to use the high efficiency solar thermoelectric generator couple with modern rainwater collection system to further increase the sustainability of the project considering Malaysia has an abundance of both of these renewable resources. The idea is to utilize the electricity generated by solar thermoelectric generator, which is a more eco-friendly and higher efficiency method of generating electricity when compared to plain solar panels production, to power the LED lights, water pumps, filtration systems and main control unit of our project to further increase the production rate of the crops and aquatic livestock. The rainwater collection system had the purpose of providing a fresh source of clean water for both the plant crops and fish while counter for the water loss due to evaporation and accidental spillage.

We also strive for the utilization of AI technologies to monitor the health and circumstances of our crops and fish to provide the best care for the quality of our final product, while at the same time minimizing any lost due to disease spreading or improper care. The AI can also be used in making decisions regarding the utilization of solar energy in a higher efficiency manner while also taking into account the day-night cycle and storage of power for any emergency use.

Part 4: Methodology

Part 4.1: Methodology for Structural Design

Our planned approach for this problem of the status quo is to design a greenhouse with IoT and AI capability. The green house will consist of three parts: Aqua the part in charge of aquafarm; Farm, the part that is related to hydroponics and Circulation, the part in charge of maintaining and monitoring the system environment and of the fish and plants that we harvest which will be assisted by the IoT system and the AI system. Hence as such our primary goal would be to create a self-warning sustainable system with the capability of identifying the errors in the system and hence form a feedback loop to ensure the ultimate sustainability of the project.

The concept of Aquaponics is by integrating Aquaculture, which is the cultivation of aquatic species such as fishes and shell fishes in a controlled environment and Hydroponics refers to the concept of growing plants in a fluid environment. The idea of the design is to use the produced waste of the fish to convert them into necessary nutrients that the plants require with the help of microbes. These microbes will extract the nitrate in the ammonia and provide them to the plants. The plants will then act as a filter by absorbing the broken-down nitrates and the water will be cleanse then return to the fish farm below the hydroponics setting. In short, aquaponics is a design that maximizes the water usage by fertilizing the crops with discharge of the fishes when in return the crops cleanse the water for the fishes.

The design of our products will be as such: The base foundation of the green house will be a 5m x 10m wide solid plastic container with anti-erosion capabilities. The depth of the foundation would be 1.5m deep and it will serve as the Aqua. On top of it is another 5m x 10m x 7.5m transparent structure built using glass which multiple PVC pipes will be attached at the side of the structure. These pipes will be served as the Farm, which is the plantation base where the water and necessary nutrients will be supplied into these pipes for the plants. The roof of the green house would be a series of solar panels connected to battery cells which allows the structure to maximize the utility of space. The pipes will be slightly rotational to ensure that the sun can shine on every piece of leaf throughout the day. Numerous sensors such as salinity, oxygen level, solar energy level sensor and other assisting devices such as microcontroller will be present in the design. Due to the fragility of the plants, another set of volt and amp meter will be placed next to each respective sensors to monitor their performance, once the sensors fail to work properly, the data will be further processed and reflected to the terminal via cloud. The exact details of the design will be further elaborated later. As such we will be able to create a complete self-monitoring network of Green House hence have the potential to generate most output in terms of aquaculture and agriculture. For our prototype, we plan to scale our design into a 1:10 model, which sizes around 0.5m x 1m x 0.9m structure

The novelty of our project lies in three unique features firstly is the concept of IoT which will allow us to maximize the efficiency of the system by preventing break downs from occurring and hence minimize the hassle needed to check on the device every day, secondly is the introduction to fermentation to maximize the utilization rate of the fish discharge by extracting most nitrates from it and thirdly is the introduction of AI recognition technology, allowing us to maximize the output of the green house by eliminating the sick or underdeveloped crops using MachineVision. The choice of crops we choose will be lettuce, spinach, and other leafy crops. The choice of fish we will be using will be catfish due to its natural advantage of requiring a lesser water quality and the choice crops is because of its adaptability in hydroponics environment.

In the current status quo, aquaponics fails to be implemented worldwide due to several reasons which includes high costs of electricity, daily maintenance, water quality difficulties and a failure of a single part will lead to breakdown of entire system (Jimi Underwood, Bruce Dunn 2017). However, in our design, we will be able to resolve all these difficulties. The cost of electricity to sustain the pumps and the uvb light will be decreased by a great extend using the presence of solar panels. We presume we will able to decrease the usage of energy by at least 20%. The daily maintenance and failure rate can be resolved using our self-monitoring IoT system which flags its flaws before they breakdown. Hence eliminating the trouble of having to monitor and maintain it every single day. Water qualities difficulties can be resolved by the water salinity sensor which will flag out the problem and cleanse the Aqua. Hence, we will able to bring the concept of aquaponics to the next level.

By doing so, out plan is to tackle multiple problems stated in the Sustainable Development Goals which are namely: Zero Hunger, Industry Innovation and Infrastructure, Sustainable cities and Communities, Responsible Consumption and Production (SGD UN, n.d.). We will be solving the Zero Hunger problem by promoting sustainable agriculture and aquaculture by maximizing the water efficiency of the system as aquaponics uses only about 10% of water consumption in soil plantation (BTL Liners, n.d.). We will be able to tackle the sector of Industry Innovation and Infrastructure, and Sustainable cities and communities by providing a sustainable infrastructure. This can be done as we use vertical farming in our design which has the capability of holding up to 665 crops in a single unit of our design compared to 95 units using soil plantation, most importantly, we will be tackling the problem of Reasonable Consumption and Production by minimizing the waste water used in aquaponics. As the water used in aquaponics will be much lesser compared to normal fish farm.

The Aqua is a solid plastic fish farm made out of High-Density Poly Ethene (HDPE) Geomembrane. The reason of choosing this type of plastic is that it is the most suitable plastic due to its high durability and cheaper price. We do acknowledge the fact that the usage of plastic is not environmentally friendly hence the choice we made has the least decomposition time compared to other alternatives. It will be partitioned into 5 sectors, the sediment tank, the fermentation tank, the purification hub, the fish pond and the backup pond. The size of Aqua is around 5meter long ad 10meter wide with 1.5 meter deep.

The sediment tank has a size of around 0.995m x 0.5m with a 1.5 meter deep. There will be two pumps in the sediment tank. One is in charge of directing the discharges of the fishes into the sedimentation tank and the other will take the responsibility of directing the sediments which has been broken down into nitrates into the next tank, which is the fermentation tank. The placing od the pumps that direct the discharge into the sediments tank will be slightly higher than the water level of the sediments tank. This is to prevent backflow of the faeces back into the fish farm and hence causing contamination to water bodies. The purpose of the fermentation tank is to allow the faeces of the fishes to be broken down into ammonium compounds such as ammonia nitrate by using bacteria such as nitrosomers and nitrobacters. These bacteria will break down the discharge of the fishes and convert them into ammonium compounds. A nitrate meter will be placed at the bottom of the tank to determine the nitrate level within the sediment tank. When the nitrate reaches a high concentrated level of around 10mg/L, the water will then be discharged into the next chamber which is the fermentation tank. The fermentation tank has a similar function and structure as the sediment tank. However instead of breaking down fish discharges into ammonia, fermentation tank breaks down ammonia compounds into nitrate fertilizers via fermentation, these fertilizers alongst with the clean water from fish farm will be directed into a tube which is connected to the plantation in Farm from the fermentation tank. Just like the sediment tank, another nitrate meter will be placed there to monitor the optimum concentration to act as the compound of the fertilizer and water. Other necessary nutrient that are premade in factory will be prepared in cases which the nutrient level in the water is lacking due to unforeseen reasons.

After the fertilizer and the water passes through the Farm, the purified water will then channel back into the filtration hub. The filtration hub serves as a buffer zone to filtrate out all the unwanted leftover nitrates. The filtration hub is connected to the main fish farm via a 0.2m diameter hole. We will be using polymer wool pad which is Aqua One Aqua Vua 28 Polymer Wool Pad, ceramic biomedia which is Marine Pure Sphere, foam of different sizes from Filson Aquarium and filter nets with the assistance of UV light steriliser (AquaMiracle Aquarium U-V Filter) to ensure that the water that is being feed into the fish farm is clean and safe for the fish to use. A pump of 300L/H with the specification type of 0Decdeal USB 5V pump will be included in the design to ensure that the water can pass through the filters effectively.

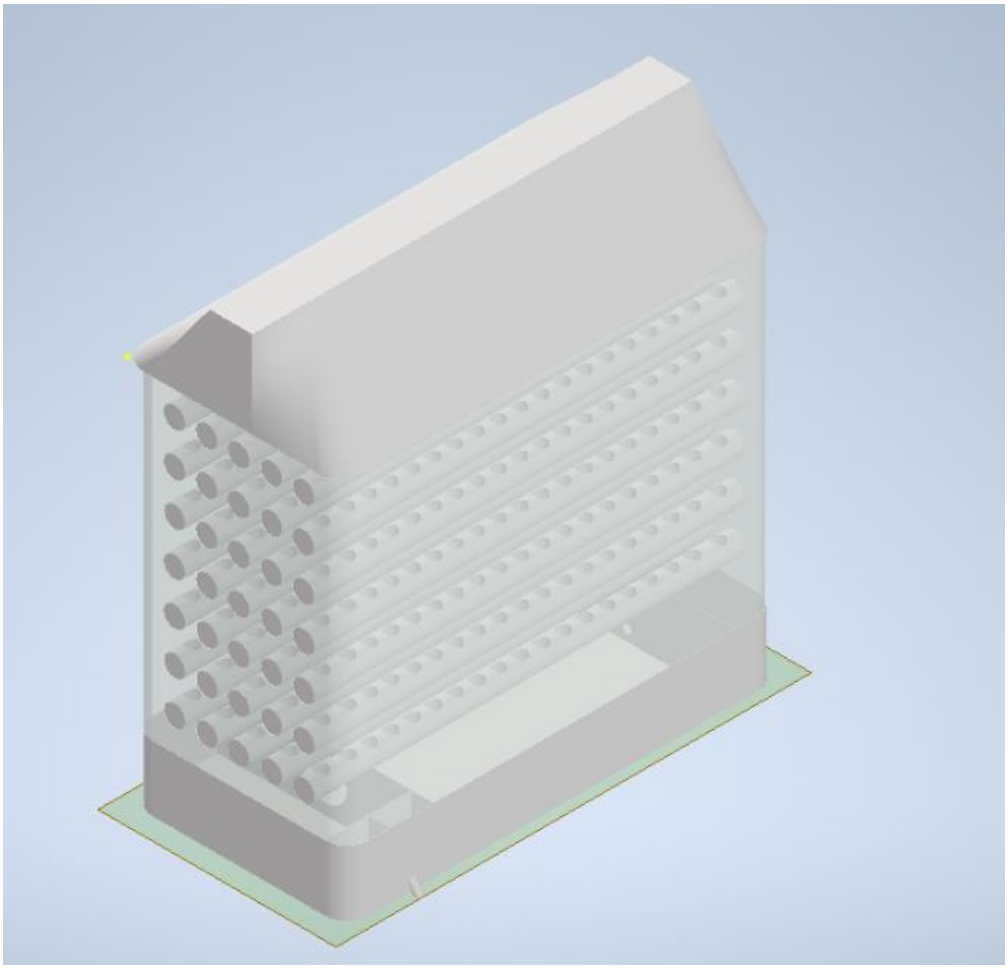
The main part of the Aqua is the fish farm, the fish farm is a 6.98m x 5m rectangle with a dept of 1.5m. The fish farm serves the purpose of allowing the fish to be cultivated there. The food will be fed at a hole next at around 1.2m height with a feeding schedule of twice a day. The estimated fish capacity is around 20 fishes in the fish farm. The amount of fish food given will be carefully calculated to ensure that there are no leftovers. Multiple sensors will be placed to monitor the water quality. For example, oximeter using a galvanic probe, salinity sensor, using a DFRobot Gravity product, temperature sensor using HTS221 and others. They will be connected to a main control device that will analyse the data and provide feedback into the system. At the same time upload the necessary data up to the cloud and into the user's application. In order to avoid fatality of the fishes due to water cleanliness, a secondary back up pond has been prepared, if the water quality is at an extremely unacceptable quality, the net that is preplaced at the back of the fish farm, directly next to the filtration hub will move and force the fishes into the backup pond through a 1-meter hole. After all the fishes has been pass through, the hole from fish farm to backup bopnd will be sealed to prevent water and fish from leaking to the fish farm which are being cleansed by completely replacing the water. The water will be discarded through the fermentation chamber, this is to maximize the waste water and convert them into ammonium compounds. After all of the water has been discarded, new, fresh water will then be filled into to the fish farm, and then the holes between fish farm and the backup pond will open up to allow the fish to flow back in. The nature of aquaponics and the presence of filtration hub ensures that the water quality will always be set at the optimum standard and hence the above mentioned procedure of having to cleanse the entire fish farm water completely and discarding them will not happen frequently, in fact this procedure merely exist to perform as a backup plan to ensure that under no circumstances that our aquaponics greenhouse will suffer a detrimental lost in fish due to unforeseen circumstances of sudden drop of quality, which is one of the disadvantages that has being brought up earlier.

The second part of our design is the Farm. The Farm in short is the hydroponics part of the structure. It consists of two parts, the circulatory system and the plantation system. By using a gap of standard hydroponics plantation, taking lettuce as an example, a gap of 0.18m should be placed between each plantation set. There are three types of aquaponics design which are Raft, NFT and media filled beds. Raft refers to the concept of the plants are grown on the boards that are usually made out of polystyrene that floats on top of the water. The second method is the nutrient film technique which is allowing a thin fil of water to flow through the roots of the plants, hence supplying nutrients for the crops. The last method is via a media filled bed which is a tank filled with gravel or other plantation bed media, the water alongst with the nutrient will then passes through the plant bed and be absorbed by the plants. In this proposal we choose the NFT method, as our design uses PVC tube with holes which the plants is gong to be placed there. Hence as PVC pipes are weak and fragile in nature, by pressuring the PVC pipe with gravels such as Media Filled Bed it might weakens the structural integrity of the Farm and hence causing danger to the daily production of Farm. Raft requires a larger amount of space for

the water to be stilled and passes through the roots of the plants, as our proposal plans to maximize the space available, it would be better for us to not chose raft as a method of aquaponics crops planting. Hence, we will be using NFT for this design.

The specific details of the design is as follows. The pump will transport the water to the PVC pipes which is served as the plantation base of the Farm, the water containing nutrients from the fermentation tank will form a thin nutrient film that will constantly flow through the plantation base. This will allow the plants to have the opportunity to absorb the nutrients that are crucial for their growth. They will then decrease the salinity improve the water cleanliness level in the process of doing so and hence able to return the water that is fully cleanse back to the filtration hub. The nutrients film contains the necessary compounds for plant growth, which mostly made out of ammonium compounds such as ammonia salt. These nutrients are widely present in the excrement of the fishes and hence widely useable for the growth of the plants.

The design of the farm will be shown as below:



The full blueprint will be attached here:
https://drive.google.com/drive/folders/1bIP_oK5Ui6IeLki3wMvMCTOb1v6Q1NM2?usp=sharing

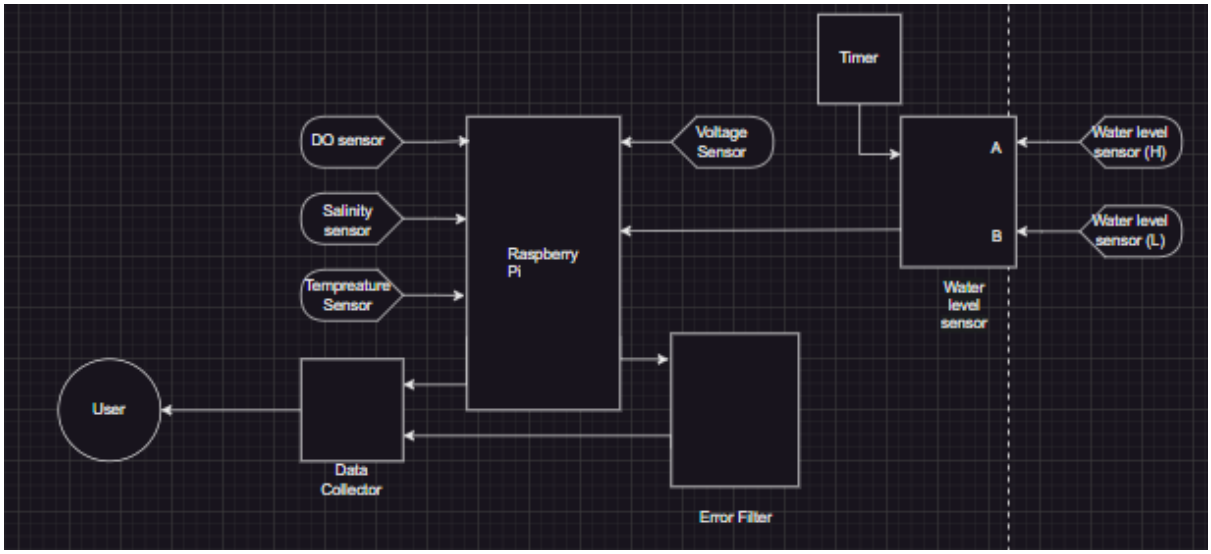
Part 4.2: Methodology for Control Unit

The Internet of Things (IoT) is used to strengthen the efficiency of monitoring can control of our systems.

First, we will be choosing Raspberry Pi Zero as the processing unit. This microcontroller is both powerful and easy to program. It's a compact and low-cost single-board computer. The variables to monitor for ensuring the optimal environment for the agricultural crops are salinity, oxygen level of the water body and the environment temperature.

A salinity sensor is used for the salinity levels of the water body, a Dissolved Oxygen sensor is used to measure the oxygen levels, and a temperature is used to measure the environment's temperature. The data is then collected and sent to the user via wireless connections. Voltage sensors are placed strategically around the circuit to prevent any error occurring. The water levels of the crops and the fishes are regulated by an automated logic control system.

The concept design for the integrated Main Control Unit (MCU) is as below:



Part 4.3: Methodology for Power Supply

For the required energy consumption of the entire Aquaponic operation, our team decided to use the Solar Thermal Energy (STE) for harnessing solar energy to generate thermal energy, which then used to power turbine engine generators for electrical energy output. When compared to conventional Solar Photovoltaic Energy (PV), STE can have an efficiency of up to 40% in terms of final conversion for electrical energy while PV only has around 15% ~ 20% efficiency. It is especially important that STE can be used as a more effective power generator in tropical countries such as Malaysia, where the average temperature is around 32 °C and sunshine hours can last up to 6 ~ 7 hours daily.

STE uses the fundamental principle of directing or reflecting sunlight into a concentrated point in space, which heats up that specific spot to a very high temperature for use in powering a boiler turbine engine. The STE produced can also be stored in high heat capacity solids such as salts or metals for later use at night without the need for storing the energy after it is converted to electrical form. The current most suitable design for our project will be the Dish Stirling type STE generator, which functions by concentrating the solar rays using mirror reflection onto a point in focal point of the concave mirrors like a satellite dish, and thereby using it as the heating element for the Stirling Engine to convert the energy into electrical energy. The entire STE structure is paired with light thermostats and controlled by the MCU described above to be always facing direct sunlight for optimal energy production.

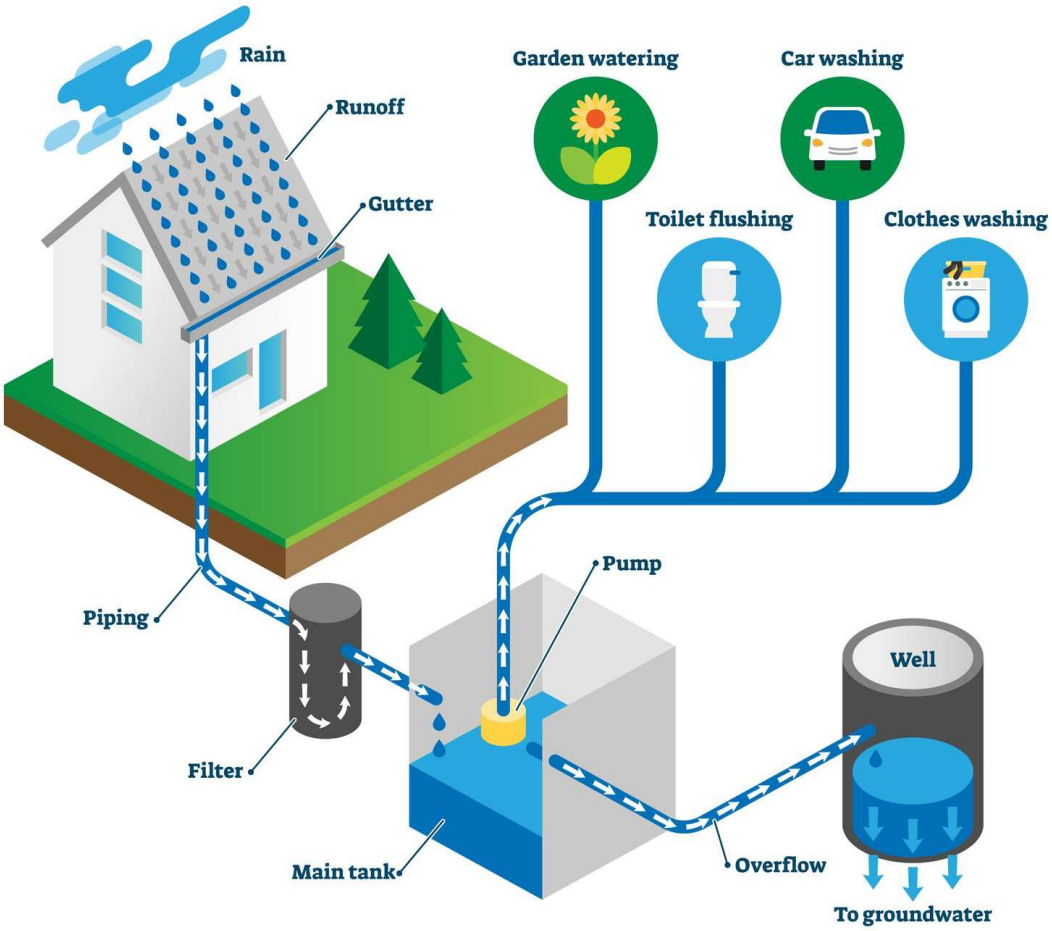


It is estimated that a Dish Stirling type STE generator such as the structure shown above with a dish diameter of around 5m can generate up to 5kWh on a sunny day while occupying less than 25 m^2 in area, with a bigger dish diameter producing an exponential growth in power generation. The generator itself is not only sufficient in powering the entire Aquaponic operation, the leftover energy can also be stored for powering the operation during hight hours where solar energy is scarce. The Dish Stirling type STE generator is also easily scalable while maintain a high build standard, allowing for larger power generation and easy maintenance.

Part 4.4: Methodology for Rainwater Collection

Rainwater harvesting is the practice of collecting and storing rain for reuse, rather than letting the water run off and be absorbed into the ground or channeled into drains, streams, or rivers. It is one of the easiest ways to conserve water for our Aquaponic operation where the rainwater can be used to replenish any water loss from the system and introduce new, fresher water for recycling.

For our project, the rainwater collection system will be placed onto the roof, where rainwater can be redirect into a storing tank for further filtering and distribution. It is important to note that Malaysia has one of the highest rainfall measured in the world, towering at 3,085.5 mm for mean annual precipitation. Such rain water can be hugely useful for watering our plants, and keeping the fishes alive and healthy, or used as cooling elements for the Dish Stirling type STE generators mentioned above.



The amount of water captured can be calculated as below:

Catchment area (roof size in square meters) ***x*** *Monthly rainfall (millimeters)* ***x*** *Conversion factor (0.62)* ***x*** *Collection factor (75%-90% to account for losses in the system)*

With the estimated roof area at around 20m^2 per module, and a monthly rainfall of 3,085.5 / 12 / 1000, the calculated water amounts to 2,710 litres of rainwater collected per month, which is way more than the water needed to be recycled into the Aquaponic system.

The rainwater collection system is also entirely functional without any energy required, which makes it effective for small scale Aquaponic operation. It is also easily scalable together with the entire project to satisfy the needs for more water into the system.

Part 5: Conclusion

We want to provide a way in which humanity can live in harmony with the nature, like how it is depicted in the concept of Solar Punk, where humans live and thrive with the help of nature's forces. We believe that Aquaponic, Solar Thermal Energy, Rainwater Harvesting and AI technology can lead humanity into a more sustainable future which is not only eco-friendly but also constructing a healthy environment for mankind to live in. The cutting edge technology and concept being implemented into our project will allow for a harmonious combination between nature and technological advancements, contriving a brighter future for all of humanity.

Part 5: Appendix and References

"How much water does aquaponics use," *www.btlliners.com*. https://www.btlliners.com/how-much-water-does-aquaponics-use#:~:text=In%20all%2C%20aquaponics%20systems%20use (accessed Mar. 14, 2024).

United Nations, "The 17 Sustainable Development Goals," *United Nations*, 2015. https://sdgs.un.org/goals (accessed Mar. 14, 2024).

J. Underwood and B. Dunn, "Aquaponics - Oklahoma State University," *extension.okstate.edu*, Feb. 2017. https://extension.okstate.edu/fact-sheets/aquaponics.html (accessed Mar. 14, 2024).

"World Bank Climate Change Knowledge Portal," *climateknowledgeportal.worldbank.org*. https://climateknowledgeportal.worldbank.org/country/malaysia/climate-data-historical#:~:text=Rainfall%20also%20remains%20high%20year (accessed Mar. 14, 2024).

"How can air pollution cause cancer?," *Cancer Research UK*, Jul. 12, 2019. https://www.cancerresearchuk.org/about-cancer/causes-of-cancer/air-pollution-

radon-gas-and-cancer/how-can-air-pollution-cause-cancer#:~:text=Because%20air%20pollution%20contains%20a (accessed Mar. 14, 2024).